Sprint 1: Requirements Engineering

TEMA: A Qualitative Analysis Application

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Section 1: Requirements

Functional Requirements

FR1 **Display Start Page:** The user should see a start page upon launching the application.
- FR1.1 **Create Canvas:** The user can create a new canvas from the home page.
- FR1.2 **View Created Canvases:** The user can open a list of previously made canvases.
  - FR1.2.1 **Open Created Canvas:** The user can select a created canvas from this list to open it within the application.
  - FR1.2.2 **Delete Created Canvas:** The user can delete a created canvases from this list.
- FR1.3 **Import Created Canvases:** The user can import a created canvas using its JSON save data.

FR2 **Send to Qualitative Analysis App:** The user must be able to send all qualitative codes to an opened canvas.
- FR2.1 **Send Google Doc Codes:** The user can send all qualitative codes within a Google Doc's comments section to an opened canvas within Google Docs.
- FR2.2 **Send Google Sheet Codes (Button):** The user can send all qualitative codes within a Google Sheet to an opened canvas within Google Sheets.
- FR2.3 **Send Google Sheets Codes (Copy Paste):** The user can copy a selection of cells in a Google Sheet and paste them into an opened canvas.

FR3 **Convert to Sticky Notes:** The system must convert each qualitative code sent via Google Doc or Google Sheets into its own sticky note.
- FR3.1 **Make User Select Canvas:** If a canvas is not opened in the application, the system shall ask the user if they want to make a new canvas for the sticky notes or to send them to an existing canvas.
- FR3.2 **Drop Sticky Notes into Canvas:** The system shall take the sticky notes gotten from Google Docs or Google Sheets and display them in the canvas in a grid pattern.
  - FR3.2.1 **Check for Existing Sticky Notes:** The system shall check for sticky notes already within the canvas to make sure the newly inserted sticky notes do not overlap with established sticky notes.

FR4 **Create Individual Sticky Notes:** The user can create an individual sticky note.

FR5 **Move Sticky Notes:** The user can drag and drop the sticky notes around the canvas.

FR6 **Edit Sticky Notes:** The user can edit a sticky note's text contents, color, size, or group(s).

FR7 **Group Sticky Notes:** The user can assign sticky notes to distinct groups.

- FR7.1 **Designate Non-Overlapping Groups:** The user can designate groups as non-overlapping. These groups cannot share sticky notes or groups with any other group.
- FR7.2 **Designate Overlapping Groups:** The user can assign groups as overlapping. These groups can share sticky notes and groups with other overlapping groups.
  - FR7.2.1 **Designate Overlapping by Default:** The system will make any newly created group overlapping by default.
- FR7.3 **Color-Code Groups:** The user can assign groups different colors. Sticky notes will change their background color based on their group's colors
  - FR7.3.1 **Color Overlapping Groups:** The system will display the background of a sticky note assigned to multiple groups as a split between these groups' colors.

FR8 **Change Sticky Note Group:** The user must be able to reassign a sticky note to different groups.

FR9 **Add Hierarchy:** The user canestablishhierarchies to the created groups, allowing for layers of subgroups within larger groups

FR10 **Change Hierarchy:** The user can change the hierarchy level of a group to a higher or lower level.

FR11 **Display Statistics:** The user can view statistics about the canvas and groups.

- FR11.1 **Display Number of Codes:** The system can show the number of codes.
  - FR11.1.1 **Display Codes per Group:** The system can show the number of codes per group.
- FR11.2 **Display Number of Groups:** The system can show the number of groups.
  - FR11.2.1 **Display Number of Subgroups per Group:** The system can show the number of subgroups within a group.
- FR11.3 **Display Dendrogram:** The system can display each hierarchy within the canvas as a dendrogram.

FR12 **Export Results:** The user can export the canvas.

- FR12.1 **Export as PNG:** The system will take a picture of the entire canvas to save as a PNG.
- FR12.2 **Export as JSON:** The system will export the canvas in a JSON format.
- FR12.3 **Export as CSV:** The system will export the canvas as a CSV.

FR13 **Refresh Data:** The user can refresh data in the canvas to see newly added codes or groups.

FR14 **Save Canvas:** The user can save their canvases.

FR15 **Assign Default Keyboard Shortcuts:** The system will have a list of predefined keyboard shortcuts that the user can use.

- FR15.1 **Change Keyboard Shortcuts:** The user can change keyboard shortcuts.

Non-Functional Requirements

NFR1	**Google Docs Code Extraction [Usability]:** The user must be able to extract codes from comments in Google Docs with one click after activating our extension.
NFR2	**Google Sheets Code Extraction [Usability]:** The user must be able to extract codes from cells within a Google Sheets with two clicks after activating our extension.
NFR3	**Displaying Codes [Responsiveness]:** The UI must remain responsive up to having 500 codes on the user's screen.
NFR4	**Deployment Mode [Portability]:** The app must run locally on a single machine.
NFR5	**In-App Sticky Notes Editing [Usability]:** Changes to sticky notes must be possible entirely within the app without opening or returning to the original Google Doc or Google Sheet.
NFR6	**Sticky Notes Editing Options [Usability]:** The user can edit sticky note properties with an average of two clicks or less.
NFR7	**Group Editing Options [Usability]:** The user must be able to assign colors for groups in at most three clicks.
NFR8	**App Theming [Flexibility]:** The user can customize the app through pre-defined themes (e.g., light mode, dark mode).
NFR9	**Keyboard Shortcut Length [Efficiency]:** Predefined shortcuts for menu options and actions (e.g., assigning a code to a group) will use no more than three keys at the same time.
NFR10	**Note Auto-Sizing [Visualization]:** Sticky notes must dynamically resize based on code text length and the number of codes within a theme.
NFR11	**Group Importance Balancing [Visualization]:** The system must provide a mechanism to balance the visual emphasis of groups based on the actual number of sticky notes they contain.
NFR12	**Confidential Data Protection [Security]:** The system must protect confidential user data.sticky notes must be possible entirely within the app without opening or returning to the original Google Doc or Google Sheet.
NFR13	**Sticky Notes Editing Options [Usability]:** The user can edit sticky note properties with an average of two clicks or less.
NFR14	**Group Editing Options [Usability]:** The user must be able to assign colors for groups in at most three clicks.
NFR15	**App Theming [Flexibility]:** The user can customize the app through pre-defined themes (e.g., light mode, dark mode).
NFR16	**Keyboard Shortcut Length [Efficiency]:** Predefined shortcuts for menu options and actions (e.g., assigning a code to a group) will use no more than three keys at the same time.
NFR17	**Note Auto-Sizing [Visualization]:** Sticky notes must dynamically resize based on code text length and the number of codes within a theme.

NFR18 **Group Importance Balancing [Visualization]:** The system must provide a mechanism to balance the visual emphasis of groups based on the actual number of sticky notes they contain.

NFR19 **Confidential Data Protection [Security]:** The system must protect confidential user data.Protection [Security]: The system must protect confidential user data.sticky notes must be possible entirely within the app without opening or returning to the original Google Doc or Google Sheet.

NFR20 **Sticky Notes Editing Options [Usability]:** The user can edit sticky note properties with an average of two clicks or less.

NFR21 **Group Editing Options [Usability]:** The user must be able to assign colors for groups in at most three clicks.

NFR22 **App Theming [Flexibility]:** The user can customize the app through pre-defined themes (e.g., light mode, dark mode).

NFR23 **Keyboard Shortcut Length [Efficiency]:** Predefined shortcuts for menu options and actions (e.g., assigning a code to a group) will use no more than three keys at the same time.

NFR24 **Note Auto-Sizing [Visualization]:** Sticky notes must dynamically resize based on code text length and the number of codes within a theme.

NFR25 **Group Importance Balancing [Visualization]:** The system must provide a mechanism to balance the visual emphasis of groups based on the actual number of sticky notes they contain.

NFR26 **Confidential Data Protection [Security]:** The system must protect confidential user data.

Constraints

C1 **Document Access [Security]:** The system must pull codes only from documents the user explicitly authenticated or grant access to.

C2 **Workspace Compatibility [Interoperability]:** The system must integrate with Google Docs and Google Sheets.

C3 **Cross Compatibility [Implementation]:** The system must be available for Linux and Windows.

Section 2: Scenarios

As-Is Scenario

Note: Thematic analysis is one type of qualitative analysis, which is then based on the coding focus (e.g., values coding)

Scenario	Setting up Codes for Thematic Analysis
Participating Actors	Goober: GraduateStudent
Flow of Events	1. Goober is a graduate student who recently conducted several long interviews with programmers on why goto is considered harmful. 2. After transcribing over ten hours of data from five participants using Google Docs, Goober then performs values coding on the transcripts. a. For values coding, Goober distills the values, attitudes and beliefs behind the participants' statements (e.g., "How did you get past security again?" is coded as "Concern about lack of security.") b. For each code, Goober highlights the portion of the transcript and writes the code as a comment. 3. After finishing the values coding of the transcripts, Goober downloads the Google Doc as an HTML file. 4. They then use a Python script sent to them by a friend to extract the comments from the HTML and save them (along with the portions the codes are assigned to) to a CSV. 5. Goober then launches Google Sheets and uploads the CSV. 6. They then open up Miroboard and create a new board. 7. Finally, they select the cells containing their codes and copy and paste them into Miroboard as sticky notes. They can now begin their thematic analysis.

Visionary Scenarios

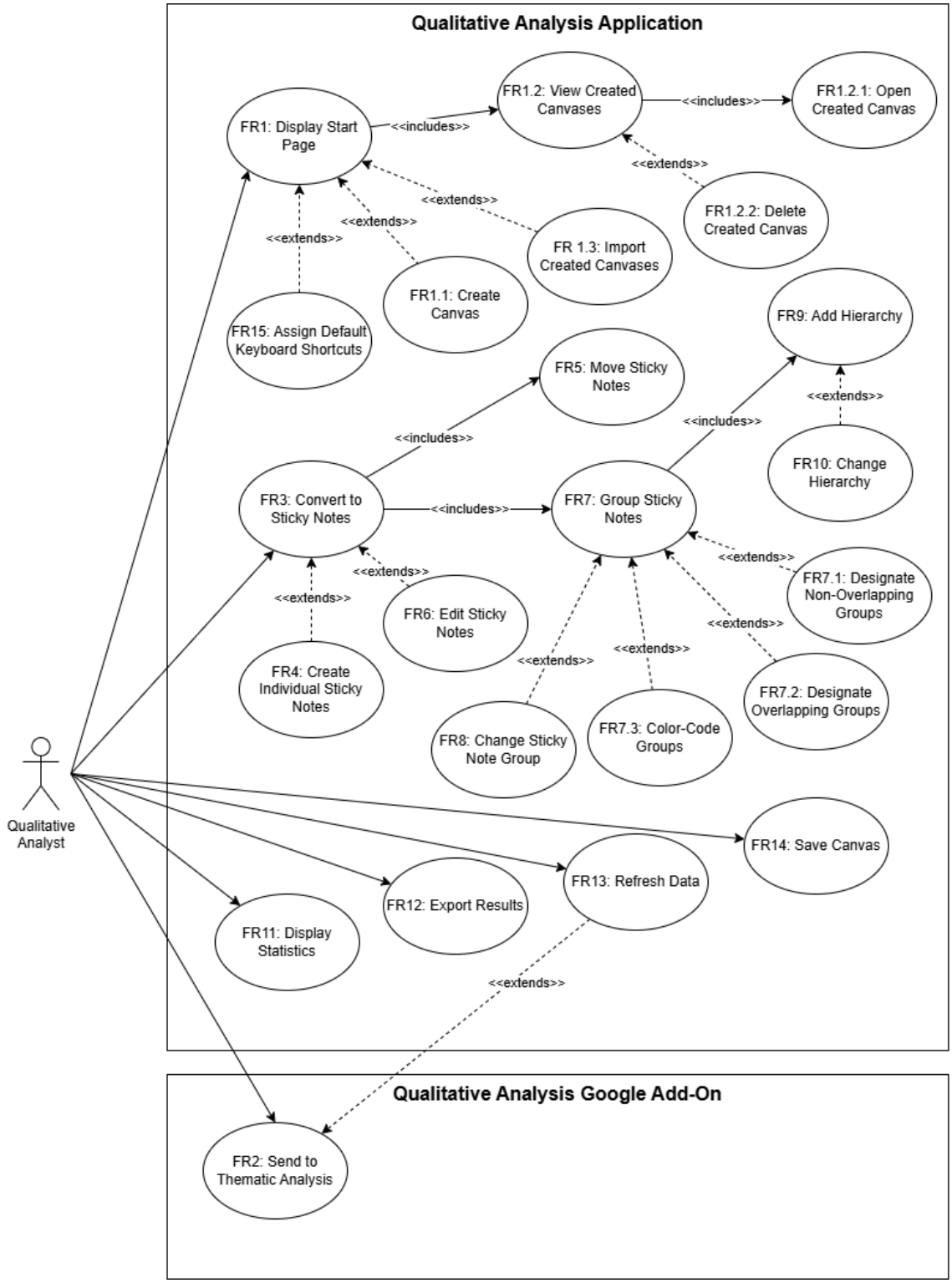
Scenario	Setting up Codes for Thematic Analysis
Participating Actors	Goober: GraduateStudent
Flow of Events	1. Goober is a graduate student who recently conducted several long interviews with programmers on why goto is considered harmful. 2. After transcribing over ten hours of data from five participants using Google Docs, Goober then performs values coding on

	the transcripts. a. For values coding, Goober distills the values, attitudes and beliefs behind the participants' statements (e.g., "How did you get past security again?" is coded as "Concern about lack of security.") b. For each code, Goober highlights the portion of the transcript and writes the code as a comment. 3. After finishing the values coding of the transcripts, Goober opens our Qualitative Analysis app. a. They create a new canvas. b. They then launch the app's extension in Google Docs. 4. Goober clicks the "export comments to the Qualitative Analysis app" button in the Google Doc add-ons. 5. The extension gathers all the relevant information (codes and associated texts) and exports them as sticky notes in the Qualitative Analysis app's opened canvas. 6. Goober switches back to the app and sees all the sticky notes ready to go. They can now begin their thematic analysis.

Scenario	Establishing a Low-Level Theme
Participating Actors	<u>Anderson: Surveyor</u>
Flow of Events	1. Anderson has just imported their codes from their large-scale survey on what people think about bread. It received over 1000 responses for the free-text questions, producing over 2000 codes, all now sticky notes. 2. Anderson has a group of 100 sticky notes all discussing how American bread is too sweet, they want to make this a low level theme. 3. Anderson navigates to the groups menu of the Qualitative Analysis app. a. Anderson clicks "create new group". b. A pop up opens where they input the group's name (American Bread Too Sweet), color (Red), level (1), and group relation (overlapping) 4. Anderson then highlights all the sticky notes which will belong to this group by dragging their mouse over the area of the canvas the codes are in. a. Anderson right clicks, and a pop up menu appears, presenting them with a list of actions for the selected sticky notes. b. Anderson selects the "assign to group action". i. A submenu appears, listing out the possible groups. ii. Anderson selects the "American Bread Too Sweet" group. 5. After selecting the group, all the sticky notes are now red. Anderson has just established a new low-level theme

Section 3: Functional Model

UML Use Case Diagram



Section 4: Reflection

Reflections

Alejandro Ciuba: Finalized FRs, NFRs, and Constraints as the stakeholder/Product Owner to better align with user needs. Developed all the scenarios (as-is and visionary).
Milan Knezevic: Contributed to initial FRs, NFRs, and Constraints definition and finalized NFRs and Constraints. Edited UML draft.
Vaibhav Singh: Contributed to initial draft of FRs, NFRs, Constraints. Built up on and finalized the UML diagram.
Daniel Yates: Contributed to initial FRs, NFRs, and Constraints definitions, and outlined the Sprint 1 deliverables document. Created draft of and finalized UML diagram, and finalized and submitted report.